6.6.6 Wrap-Up: Reflection

1. Draw an **X** on each line to indicate how you feel about each learning target.

    a. I can graph the solution set of a real-world linear inequality, given in standard form, slope-intercept form, or point-slope form on a coordinate plane.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

    b. I can interpret solutions to problems as viable or non-viable options.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

Unit 6, Lesson 7: Writing Two-Variable Linear Inequalities

Benchmark

MA.912.AR.2.7 Write two-variable linear inequalities to represent relationships between quantities from a graph or a written description within a mathematical or real-world context.

Learning Target

☐ I can write a two-variable inequality given the graph of a solution set.

6.7.1 Warm-Up: Which One Doesn’t Belong?

1. Four figures are shown.

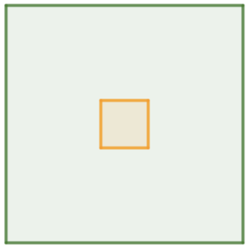


Figure 1

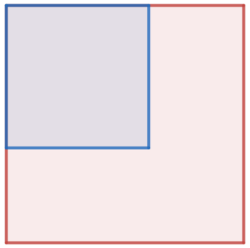


Figure 2

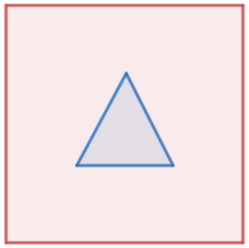


Figure 3

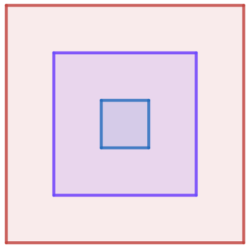


Figure 4

a. Which one doesn’t belong?

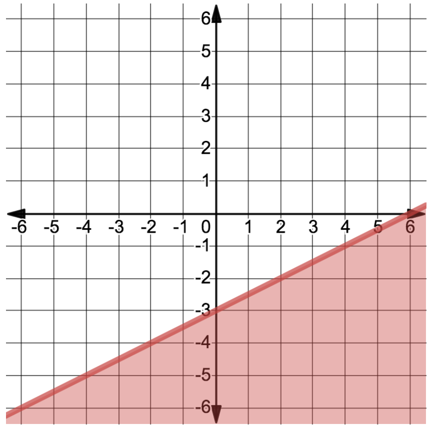
b. Explain your choice.

6.7.2 Exploration: Relating Linear Equations to Linear Inequalities

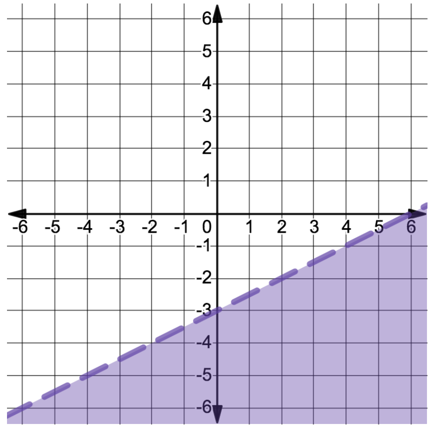
1. Four inequalities are given.

$y < \frac{1}{2}x - 3$ $\quad$ $y + 4 > \frac{1}{2}(x + 2)$ $\quad$ $y \leq 2x - 3$ $\quad$ $-x + 2y \leq -6$

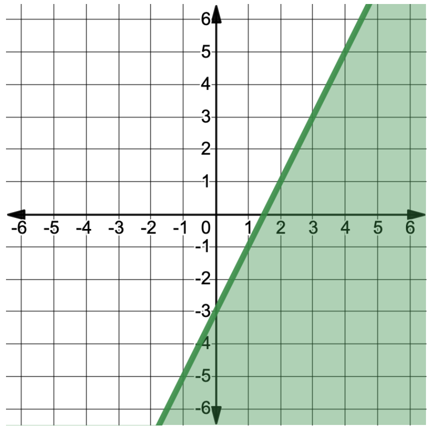
a. Work with your partner to match each inequality to the graph of its solution set by writing the inequality on the line provided under each graph.



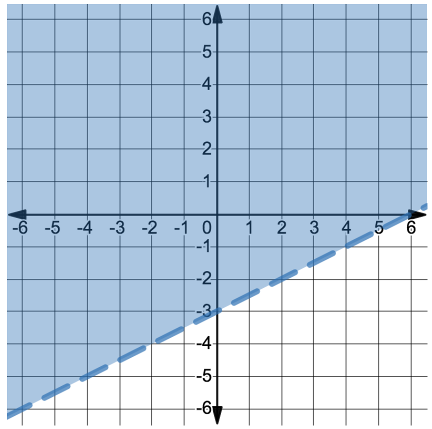
Inequality: ______________



Inequality: ______________



Inequality: ______________



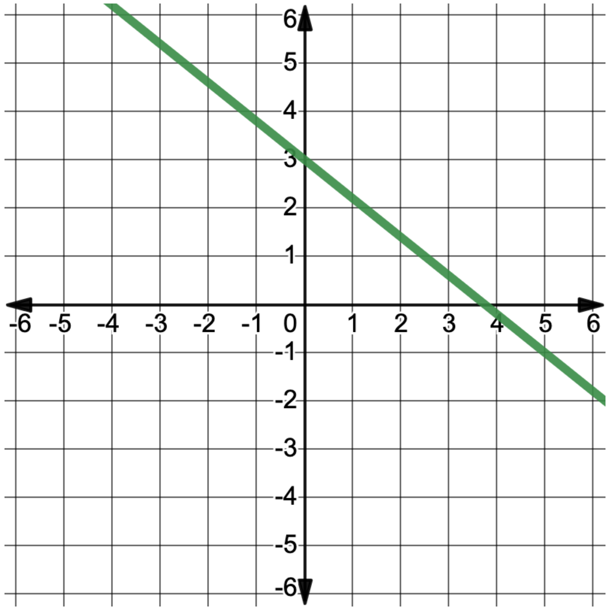
Inequality: ______________

b. Explain which graph(s) were the easiest for you to match the graph to the inequality.

c. Which of the inequalities have the same boundary line?

d. Explain how you distinguished between the graphs of the inequalities with the same boundary line using the different forms of the inequalities given.

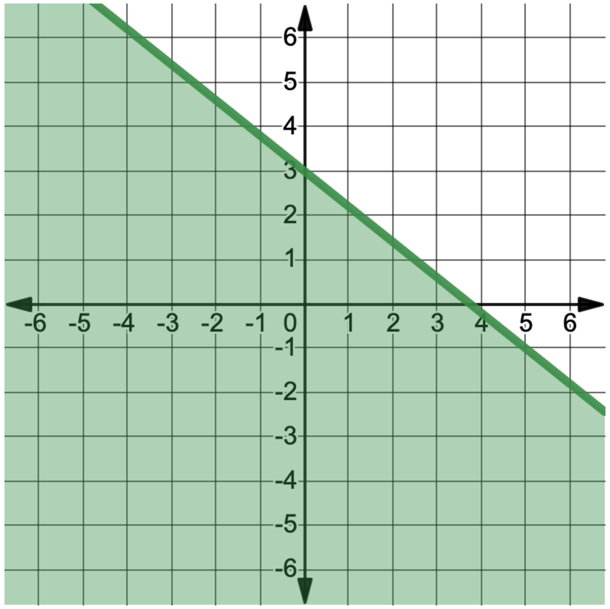
2. The graph of a two-variable linear equation is shown.



a. Write the equation of the line in slope-intercept form.

b. Suppose the line shown on the coordinate grid is the boundary line of a two-variable linear inequality. Discuss with your partner which parts of the inequality would be the same as the equation from Part A, and which parts of the inequality would be different.

c. The graph shows the solution set of an inequality with the same line as the boundary line.

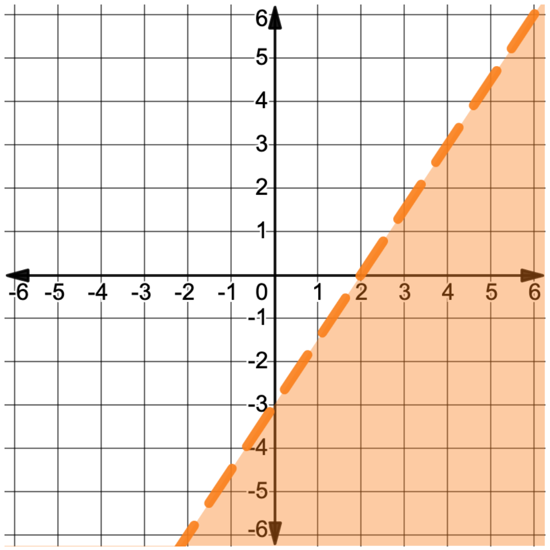


Write the inequality represented on the graph in slope-intercept form.

6.7.3 Guided Instruction: Writing Two-Variable Linear Inequalities in Different Forms

Similar to the equations of lines, two-variable linear inequalities can be written in slope-intercept, point-slope, or standard form.

1. The graph of a two-variable linear inequality is shown.



   a. Write the inequality of the solution set graphed above in **slope-intercept form**.

   b. Explain how to determine which inequality symbol to use when writing an inequality in slope-intercept form.

2. Consider how to use the graph to write the inequality in **point-slope form**.

   a. Write an equation for the boundary line in point-slope form.

   b. Select a test point on one side of the boundary line. Substitute the x- and y-value of the test point into the equation of the boundary line in point-slope form and evaluate. Show your work.

   c. Use your work from Part B to determine which inequality symbol to use. Then, write an inequality in point-slope form for the graphed solution set.

   d. Explain why there are many different point-slope form inequalities that could have the same solution set.

3. Eli and Mariska each wrote an inequality in **standard form** for the same graphed solution set. Their inequalities and reasoning are shown.

Eli	Mariska
"I took the slope-intercept form inequality and re-wrote it in standard form. First, I multiplied the inequality by 2 to eliminate the fractional coefficient, and then I added $-3x$ to both sides of the inequality as shown." $y < \frac{3}{2}x - 3$ $2y < 3x - 6$ $-3x + 2y < -6$	"I looked at the intercepts of the boundary line, $(0, -3)$ and $(2, 0)$. I know that 18 is a common multiple of -3 and 2, so I wrote an equation of the boundary line set equal to 18. Then, I used the intercepts to determine the coefficients of each term in the equation $9x - 6y = 18$. Lastly, I used a test point to determine the inequality symbol to use, and I got $9x - 6y > 18$."

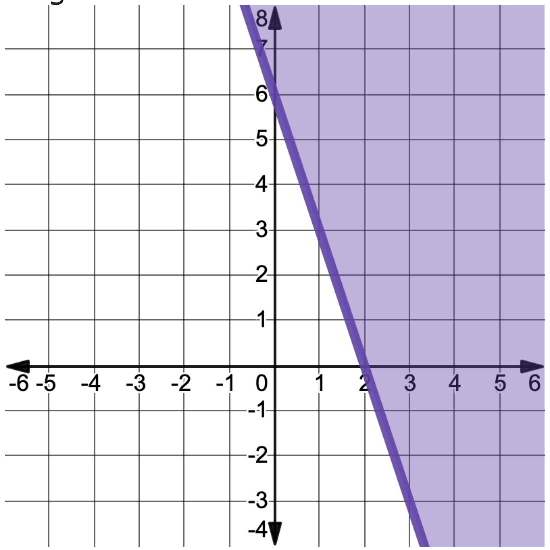
a. Discuss with your partner whether each student is correct or incorrect. Summarize your discussion.

b. Write another inequality in standard form that would have the same solution set.

Given the graph of a solution set, there are **infinitely many** inequalities that could be written with that solution set. The inequality can be written in slope-intercept form, point-slope form, or standard form.

6.7.4 Guided Practice: Writing Inequalities Given the Graph of the Solution Set

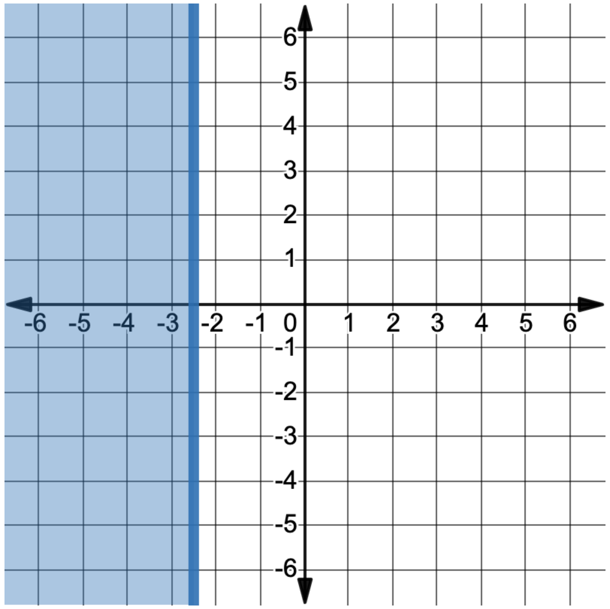
1. The solution set of an inequality is shown on the coordinate grid.



Write an inequality in each form given the solution set shown on the graph. Check your work by substituting a test point into each inequality.

	Inequality	Check with Test Point
Slope-Intercept Form		
Point-Slope Form		
Standard Form		

2. The solution set of an inequality is graphed.



a. Which of the following inequalities represents the solution set graphed above?

- ○ $y \geq -2.5$
- ○ $y \leq -2.5$
- ○ $x \geq -2.5$
- ○ $x \leq -2.5$

b. Explain your selection.

6.7.5 Activity: Scavenger Hunt

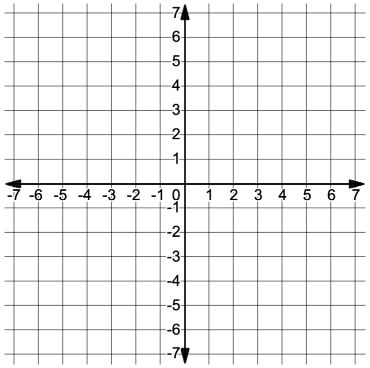
Work with your partner or group to complete the scavenger hunt. Each poster shows the following information.

Poster ____	
Inequality from Previous Poster:	*(The inequality you wrote from the previous poster will appear here.)*

Result from previous problem

Write an inequality in ________________ **form** with the solution set shown.

New Problem



- Write the letter of the poster you start with in the first row of the table. Then, work with your group or partner to write an inequality for the graphed solution set shown on the poster in the form indicated. Verify that your inequality correctly matches the solution set using a test point.

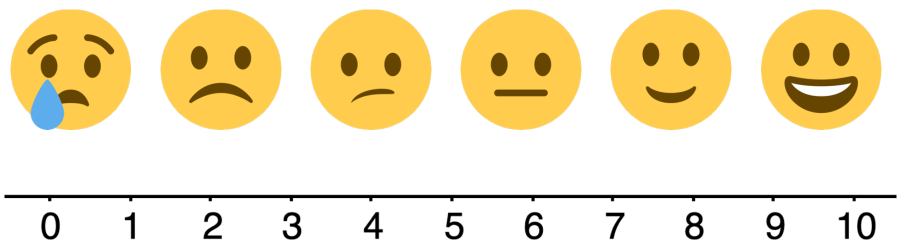
- Look for the card that matches your inequality in the grey area to find your next problem.

- Repeat this process until your group has completed all the scavenger hunt problems and end up back at your original problem.

Poster	Inequality in the Form Indicated	Verify With a Test Point

6.7.6 Wrap-Up: Reflection

1. Use the emoji scale to describe how confident you feel in writing a two-variable inequality in different forms when given the graph of a solution set.



2. Explain your selection.

Unit 6, Lesson 8: Writing a Two-Variable Linear Inequality to Represent a Real-World Context

Benchmarks

MA.912.AR.2.7 Write two-variable linear inequalities to represent relationships between quantities from a graph or a written description within a mathematical or real-world context.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.

Learning Targets

- ☐ I can write a linear inequality given a real-world context.
- ☐ I can interpret solutions to problems as viable or non-viable options.